ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

# MATHEMATICS

## PRACTICE EXAM 1 · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 8 Mathematics practice exam 1.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

## QUESTION 1

**1** 1.1 D ✓ 1.2 C ✓ 1.3 A ✓ 1.4 B ✓ 1.5 B ✓ 1.6 B ✓ 1.7 B ✓ 1.8 B ✓ 1.9 C ✓ 1.10 A ✓ [10]

## QUESTION 2

**2.1** $240 = 2^4 \times 3 \times 5$ ✓ $360 = 2^3 \times 3^2 \times 5$ ✓ (4)
$\text{HCF} = 2^3 \times 3 \times 5 = 120$ ✓ $\text{LCM} = 2^4 \times 3^2 \times 5 = 720$ ✓

**2.2** $\text{SI} = 8\,000 \times 0{,}09 \times 4$ ✓ $= \text{R}2\,880$ ✓ $\text{Total} = \text{R}10\,880$ ✓ (3)

**2.3.1** $10\% \times \text{R}6\,500 = \text{R}650$ ✓ (1)

**2.3.2** $18 \times \text{R}390 = \text{R}7\,020$ ✓ $\text{R}650 + \text{R}7\,020 = \text{R}7\,670$ ✓ (2)

**2.3.3** $\text{R}7\,670 - \text{R}6\,500 = \text{R}1\,170$ ✓ (1)

**[11]**

## QUESTION 3

**3.1** (a) $40 + (-3)$ ✓ $= 37$ ✓ (6)
(b) $16 - 9$ ✓ $= 7$ ✓
(c) $-4 + 7$ ✓ $= 3$ ✓

**3.2** (a) $4a^6$ ✓✓ (4 ✓, $a^6$ ✓) (b) $5x^4$ ✓ (c) $1 + 1 = 2$ ✓ (4)

**[10]**

## QUESTION 4

**4.1.1** ${}^7/_3 + {}^7/_4$ ✓ $= {}^{28}/_{12} + {}^{21}/_{12}$ ✓ $= {}^{49}/_{12} = 4\,{}^1/_{12}$ ✓ (3)

**4.1.2** ${}^3/_5 \times {}^{10}/_9$ ✓ $= {}^{30}/_{45} = {}^2/_3$ ✓ (2)

**4.1.3** ${}^4/_9 - {}^1/_9$ ✓ $= {}^3/_9 = {}^1/_3$ ✓ (2)

**4.2** (a) 0,3 ✓ (b) $720 \div 9$ ✓ $= 80$ ✓ (3)

**[10]**

## QUESTION 5

**5.1.1** $\text{Constant difference} = 3$ ✓ $\rightarrow \text{T}_n = 3n + 2$ ✓ (2)

**5.1.2** $\text{T}_{25} = 3(25) + 2 = 77$ ✓ (1)

**5.1.3** $3n + 2 = 152 \rightarrow 3n = 150$ ✓ $\rightarrow n = 50$; it is the 50th term ✓ (2)

**5.2** (a) $-2(-2) + 1 = 5$ ✓ (b) $-2(2) + 1 = -3$ ✓ (c) $-2p + 1 = -9 \rightarrow p = 5$ ✓ (3)

**5.3** (a) $5x^2$ ✓ $+ 2x$ ✓ (6)
(b) $-6x^2$ ✓ $+ 8x$ ✓
(c) $3a^2$ ✓ $- 2a$ ✓

**5.4** (a) $3x = -15$ ✓ $\rightarrow x = -5$ ✓ (5)
(b) $x - 3 = 7$ (or $2x - 6 = 14$) ✓ $\rightarrow x = 10$ ✓
(c) $3^x = 3^4 \rightarrow x = 4$ ✓

**5.5** $3x - 7 = x + 9 \rightarrow 2x = 16 \rightarrow x = 8$ ✓ (1)

**[20]**

## QUESTION 6

**6.1** (a) $\angle GHD = 65°$ ✓ (corresponding angles, AB ∥ CD) ✓ (6)
(b) $\angle BGH = 180° - 65° = 115°$ ✓ (adjacent angles on a straight line) ✓
(c) $\angle GHC = 180° - 65° = 115°$ ✓ (adjacent angles on a straight line; or alternate angles with $\angle BGH$, AB ∥ CD) ✓

**6.2** $2x + 3x + 4x = 180°$ (sum of the angles of a triangle) ✓ $\rightarrow 9x = 180° \rightarrow x = 20°$ ✓ (3)
$\angle P = 40°, \angle Q = 60°, \angle R = 80°$ ✓

**6.3** $\angle KMN = 48° + 67° = 115°$ ✓ (the exterior angle of a triangle equals the sum of the two interior opposite angles) ✓ (2)

**6.4** (a) SSS ✓ (b) SAS ✓ (2)

**[13]**

## QUESTION 7

**7.1** $h^2 = 5^2 - 1{,}4^2$ ✓ $= 25 - 1{,}96 = 23{,}04$ ✓ $\rightarrow h = 4{,}8$ m ✓ (3)

**7.2** (a) $A = \pi r^2 = {}^{22}/_{7} \times 14 \times 14$ ✓ $= 616$ cm² ✓ (4)
(b) $C = 2\pi r = 2 \times {}^{22}/_{7} \times 14$ ✓ $= 88$ cm ✓

**7.3** (a) V = area of cross-section × length $= ½ \times 3 \times 4 \times 10$ ✓ $= 60$ cm³ ✓ (5)
(b) SA $= 2(½ \times 3 \times 4) + (3 + 4 + 5) \times 10$ ✓ $= 12 + 120$ ✓ $= 132$ cm² ✓

**[12]**

## QUESTION 8

**8.1** 8.1.1 $(3\ ;\ 2)$ ✓ 8.1.2 $(-3\ ;\ -2)$ ✓ 8.1.3 $(-1\ ;\ 1)$ ✓✓ (one mark per coordinate) 8.1.4 $(-2\ ;\ 3)$ ✓ 8.1.5 $(6\ ;\ -4)$ ✓ (6)

**[6]**

## QUESTION 9

**9.1.1** $600 \div 10$ ✓ $= 60$ ✓ (2)

**9.1.2** Ordered: $39\ ;\ 45\ ;\ 55\ ;\ 58\ ;\ 62\ ;\ 62\ ;\ 62\ ;\ 66\ ;\ 71\ ;\ 80$ ✓ $\rightarrow$ median $= (62 + 62) \div 2 = 62$ ✓ (2)

**9.1.3** $62$ ✓ (1)

**9.1.4** $80 - 39 = 41$ ✓ (1)

**9.2** (a) $^{12}/_{60} = {}^{1}/_{5}$ ✓ (b) $^{1}/_{6}$ ✓ (2)

**[8]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**